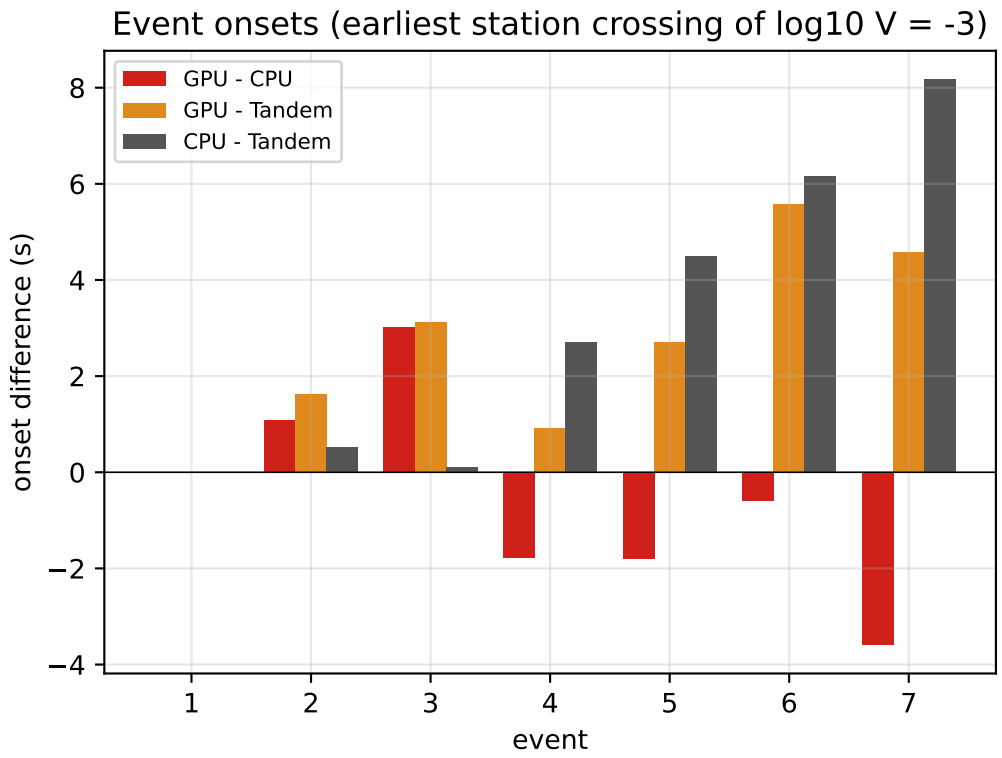
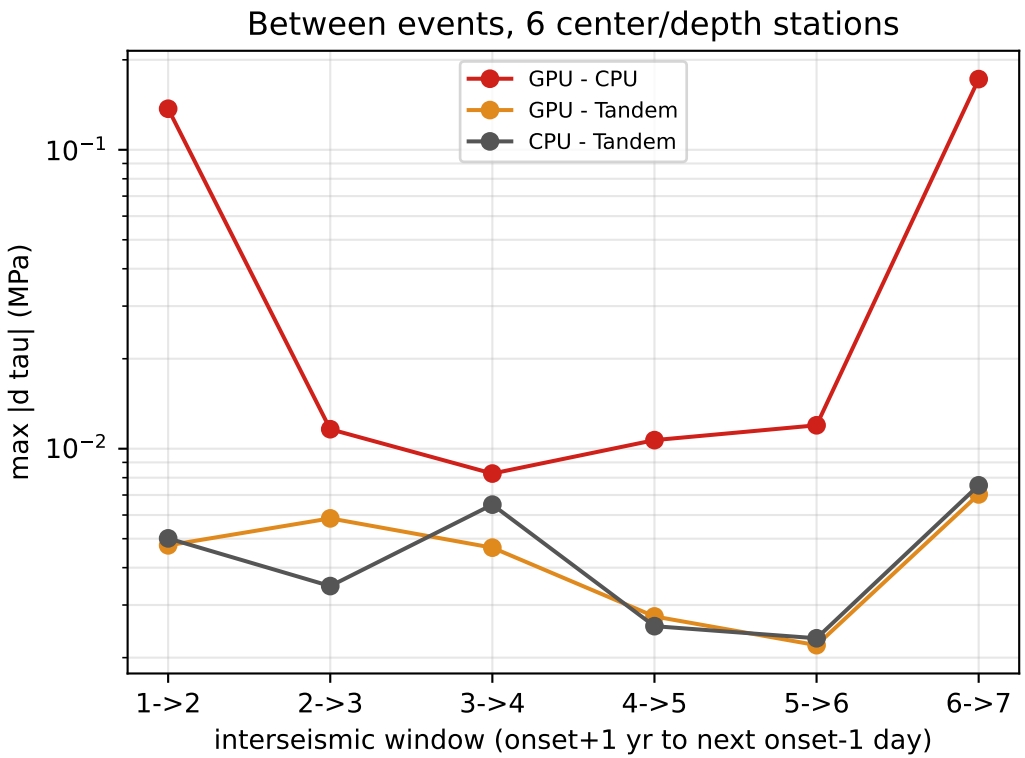
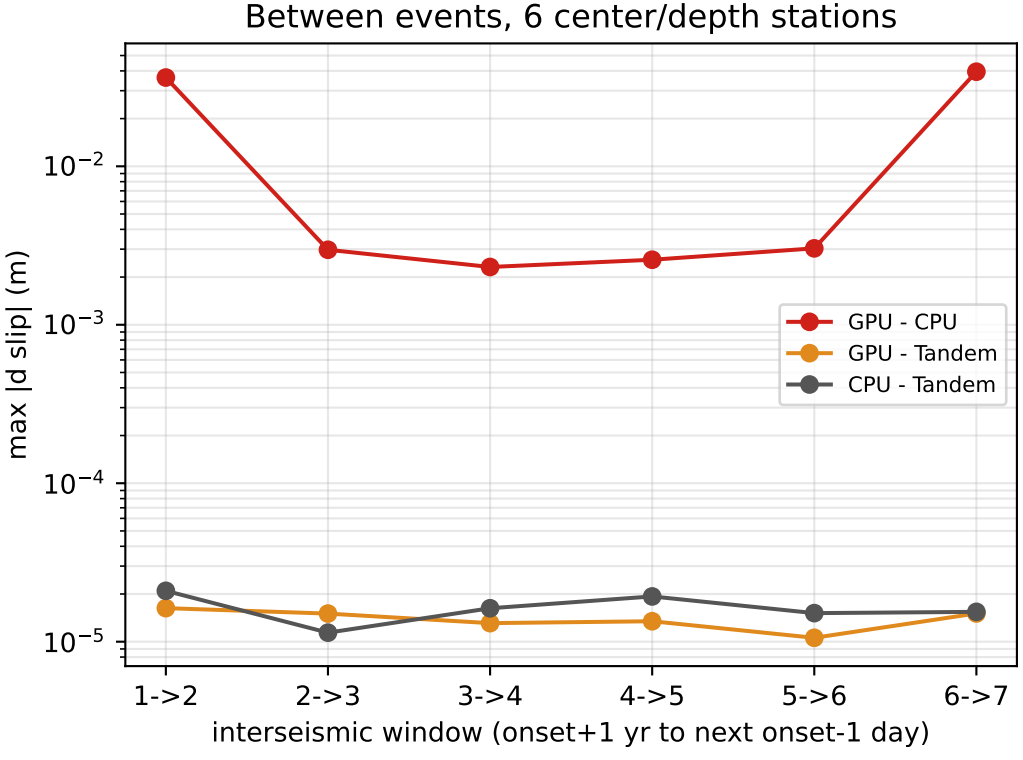
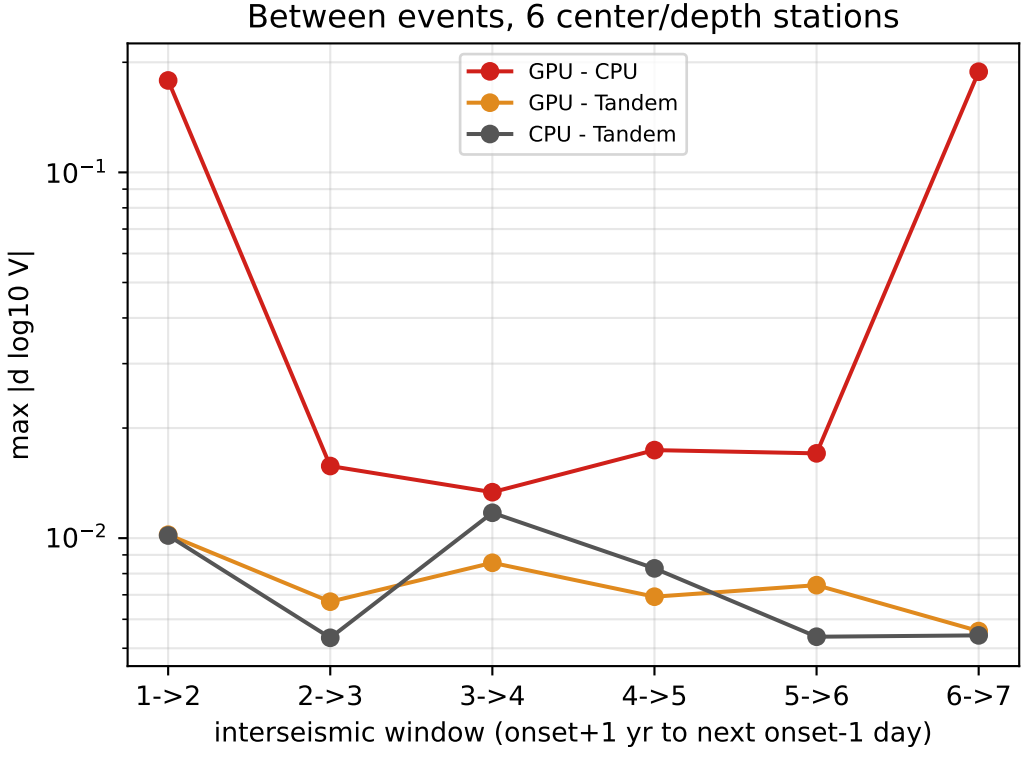
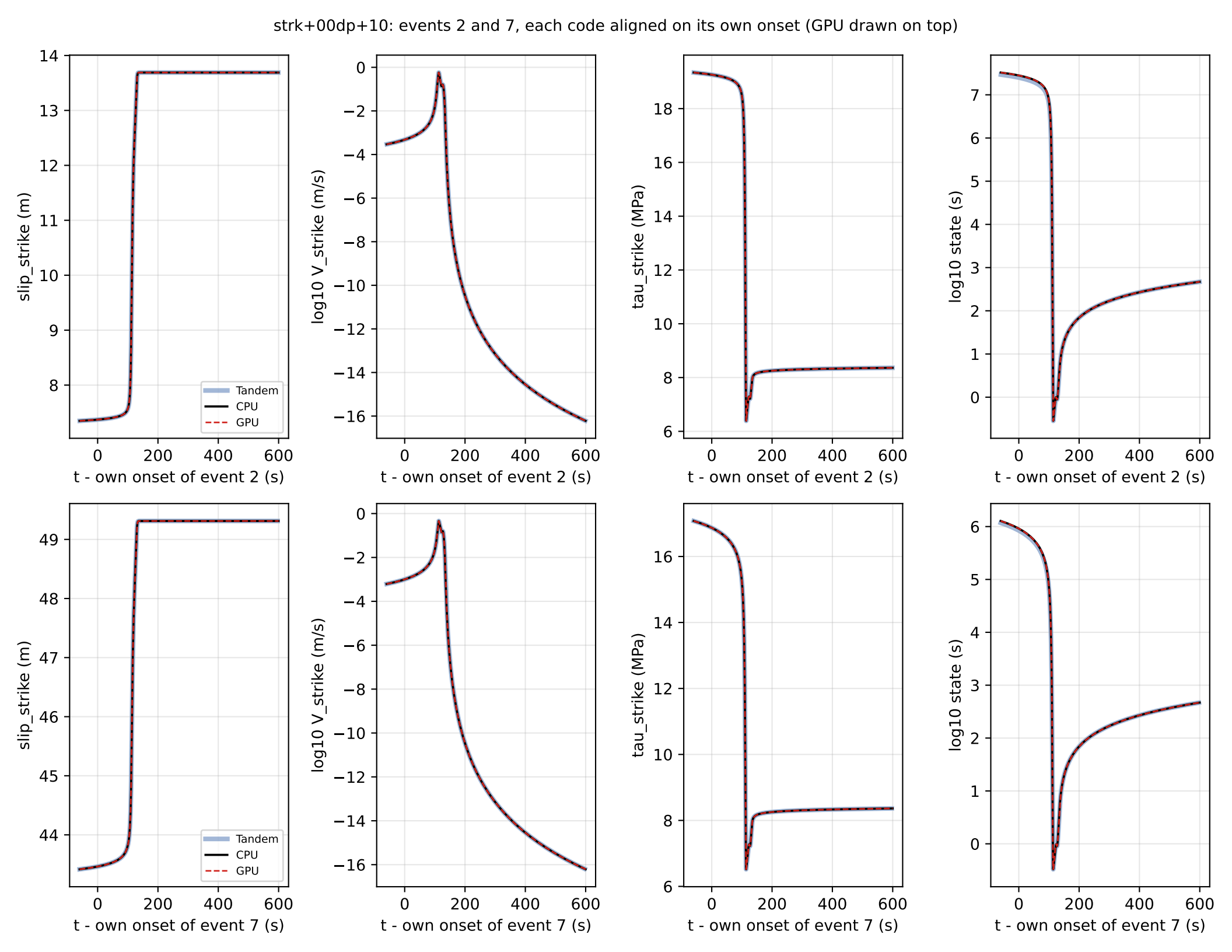
BP5-QD 1000 m p=1: GPU 69011 vs CPU 1074099 vs Tandem p=1 (compare_tandem_p1 method, 365.25-day years)

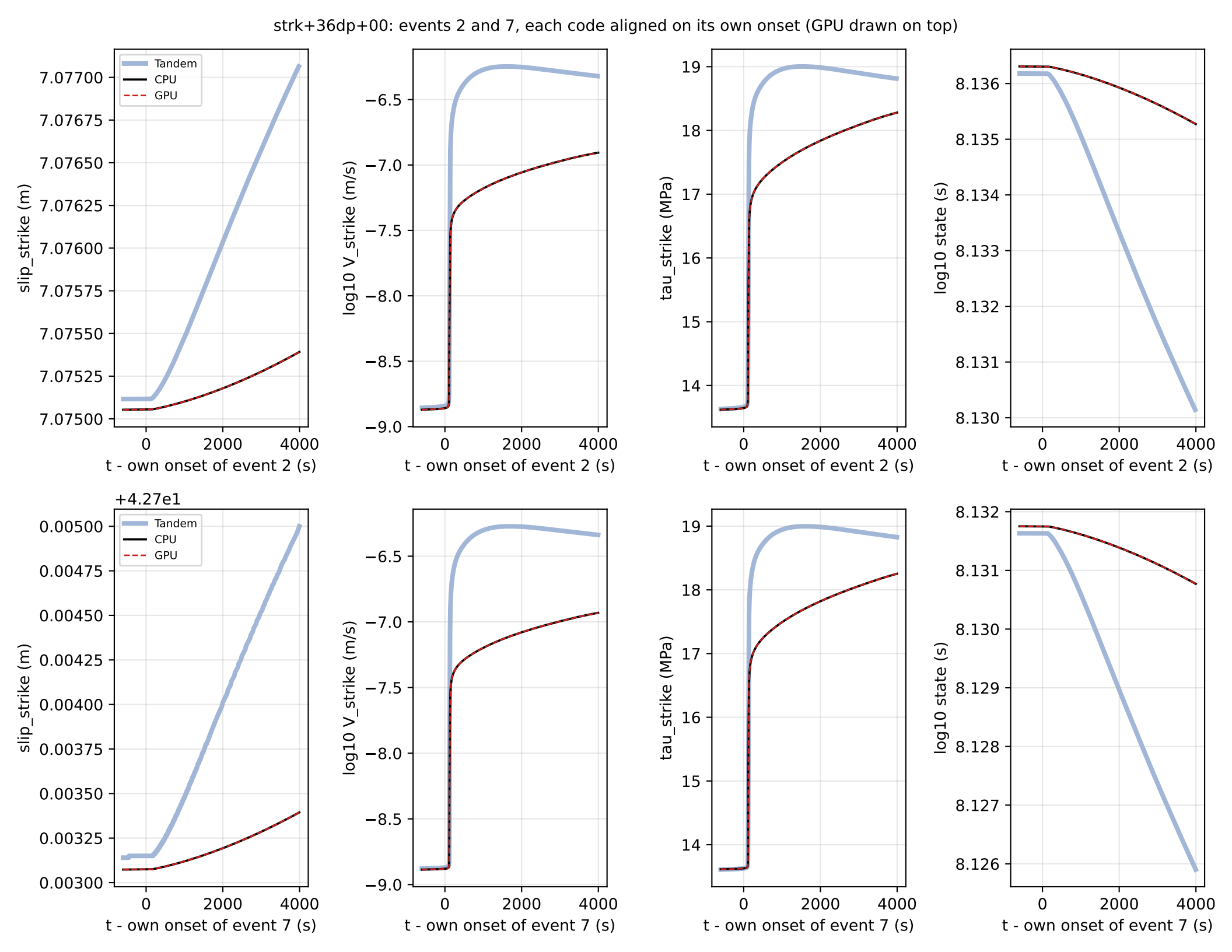


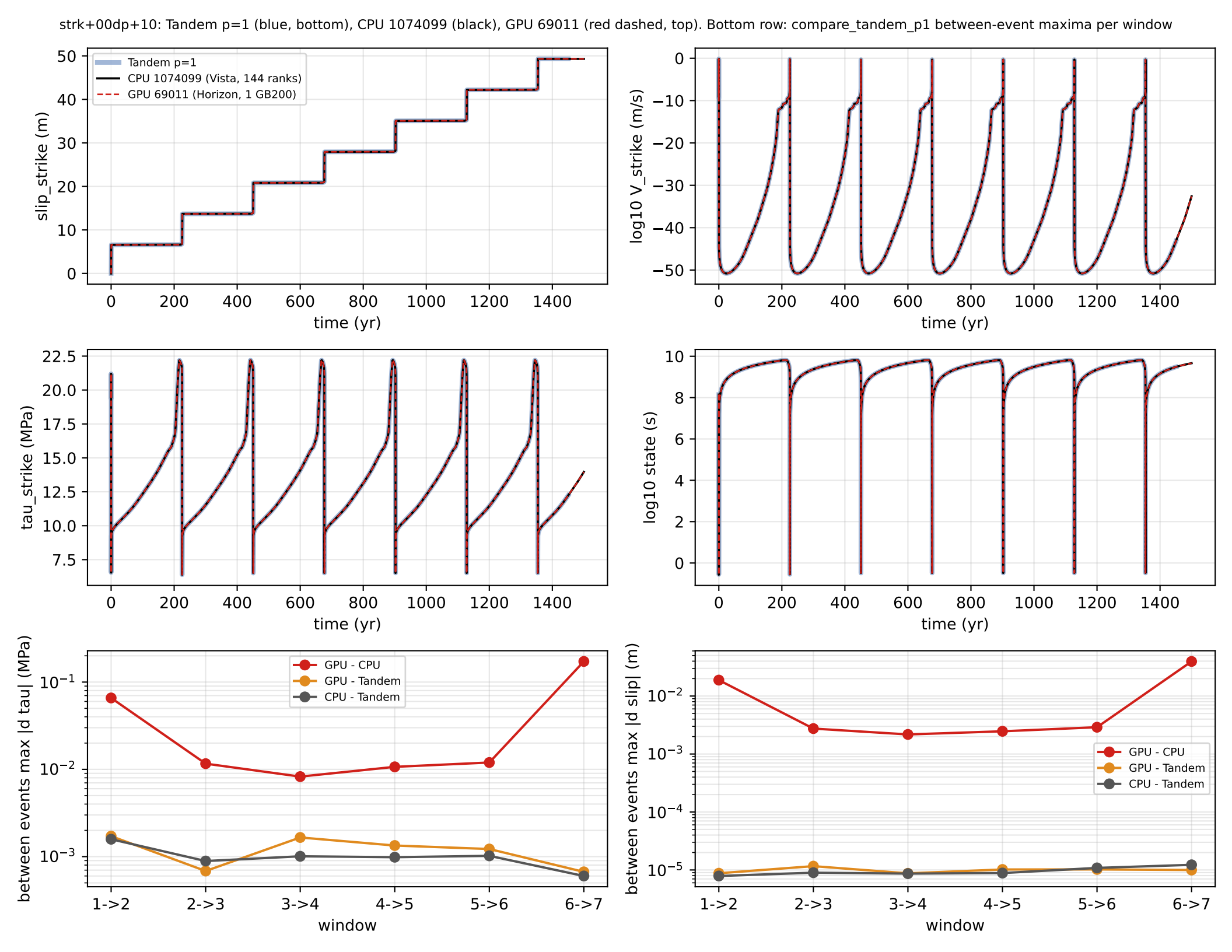


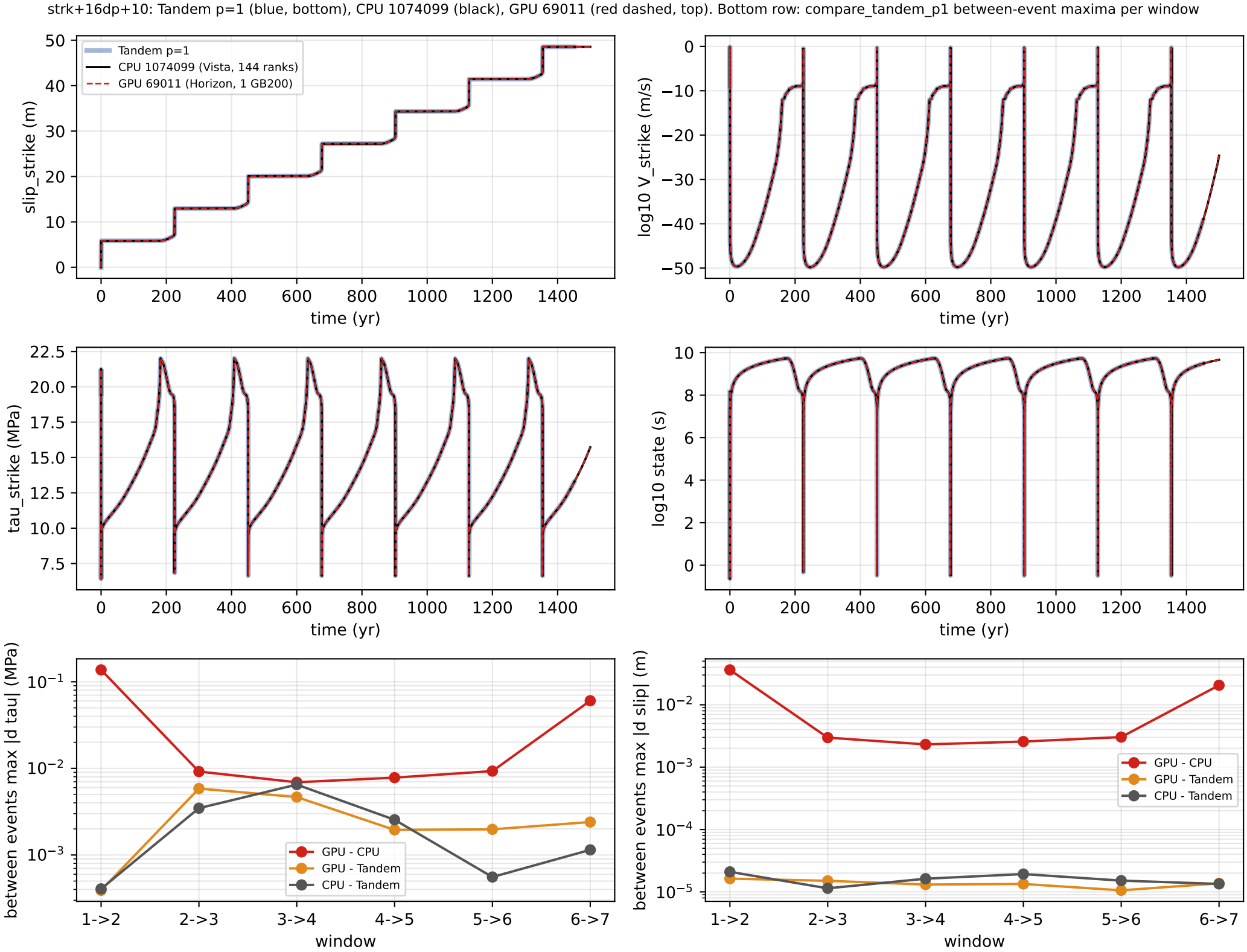


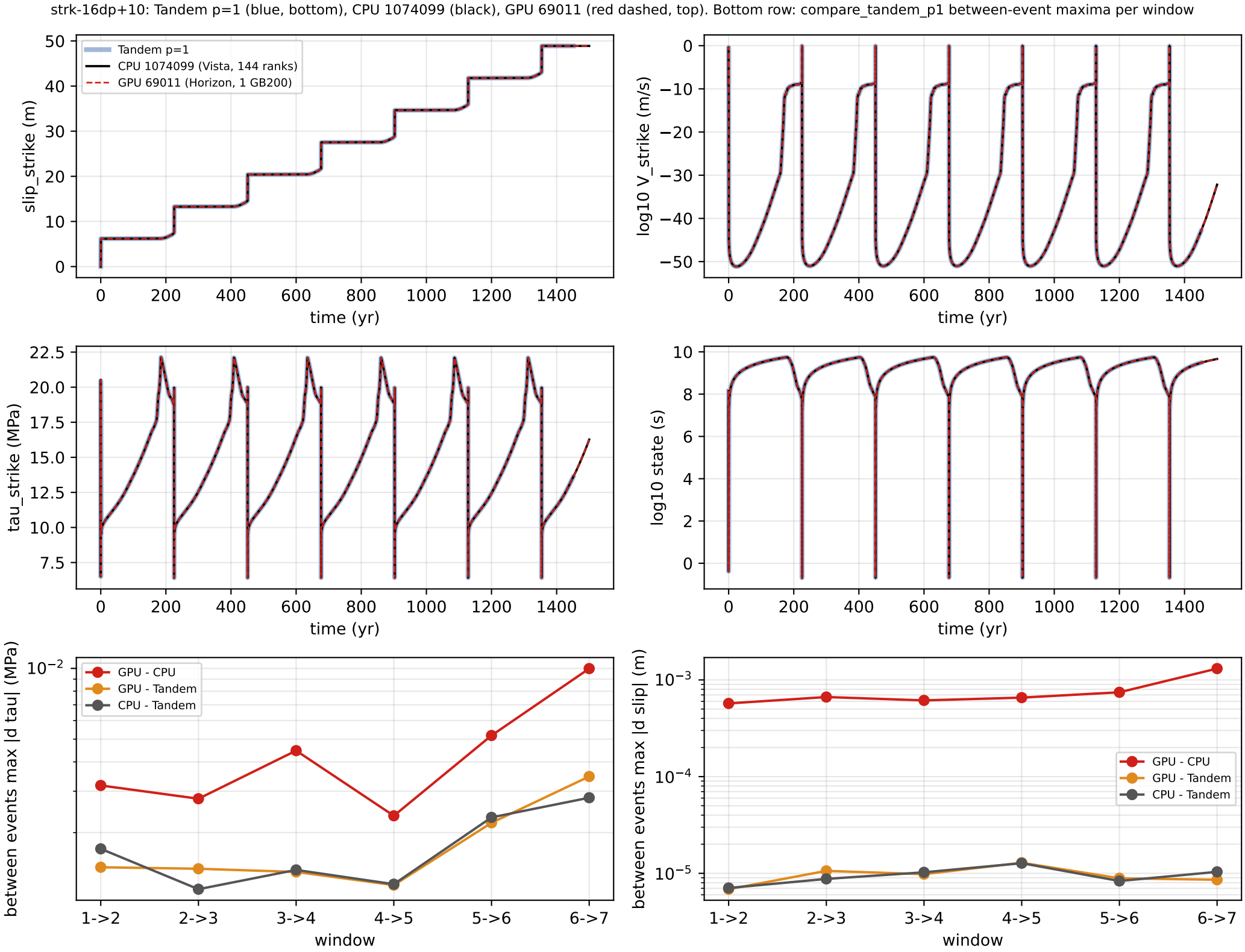


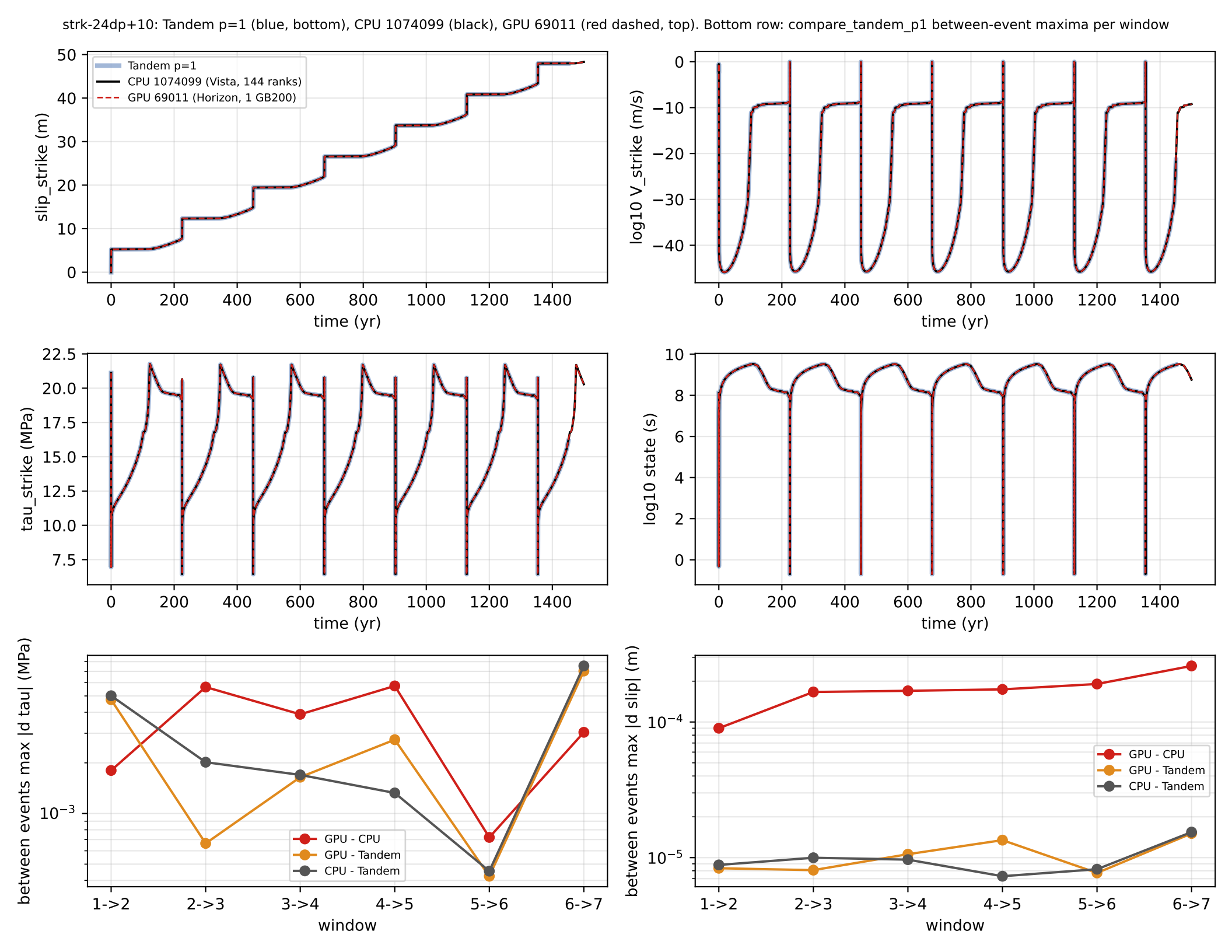

strk+00dp+10: events 2 and 7, each code aligned on its own onset (GPU drawn on top)
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
t - own onset of event 2 (s)
t - own onset of event 7 (s)
Tandem
CPU
GPU

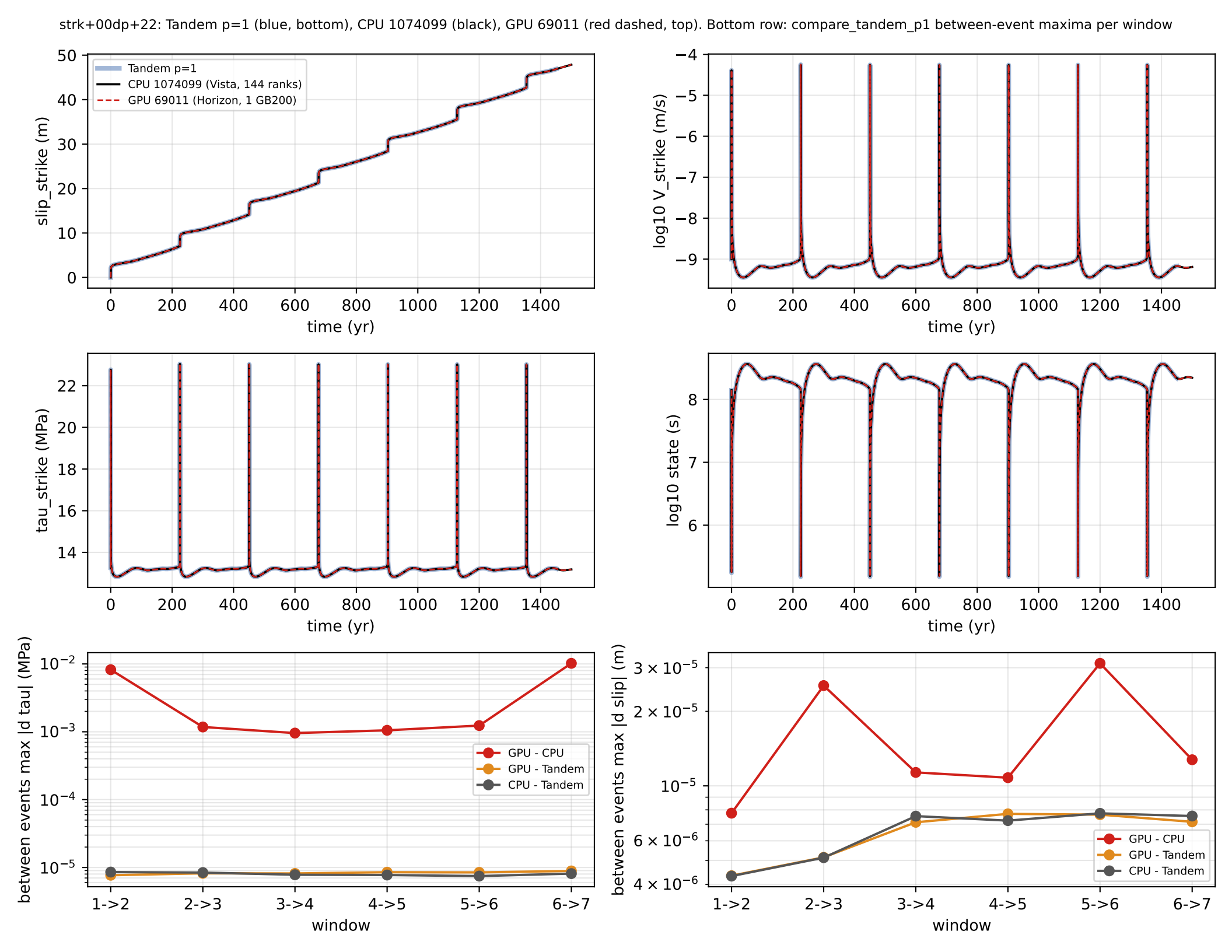

strk+36dp+00: events 2 and 7, each code aligned on its own onset (GPU drawn on top)
Tandem
CPU
GPU
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
t - own onset of event 2 (s)
+4.27e1
t - own onset of event 7 (s)

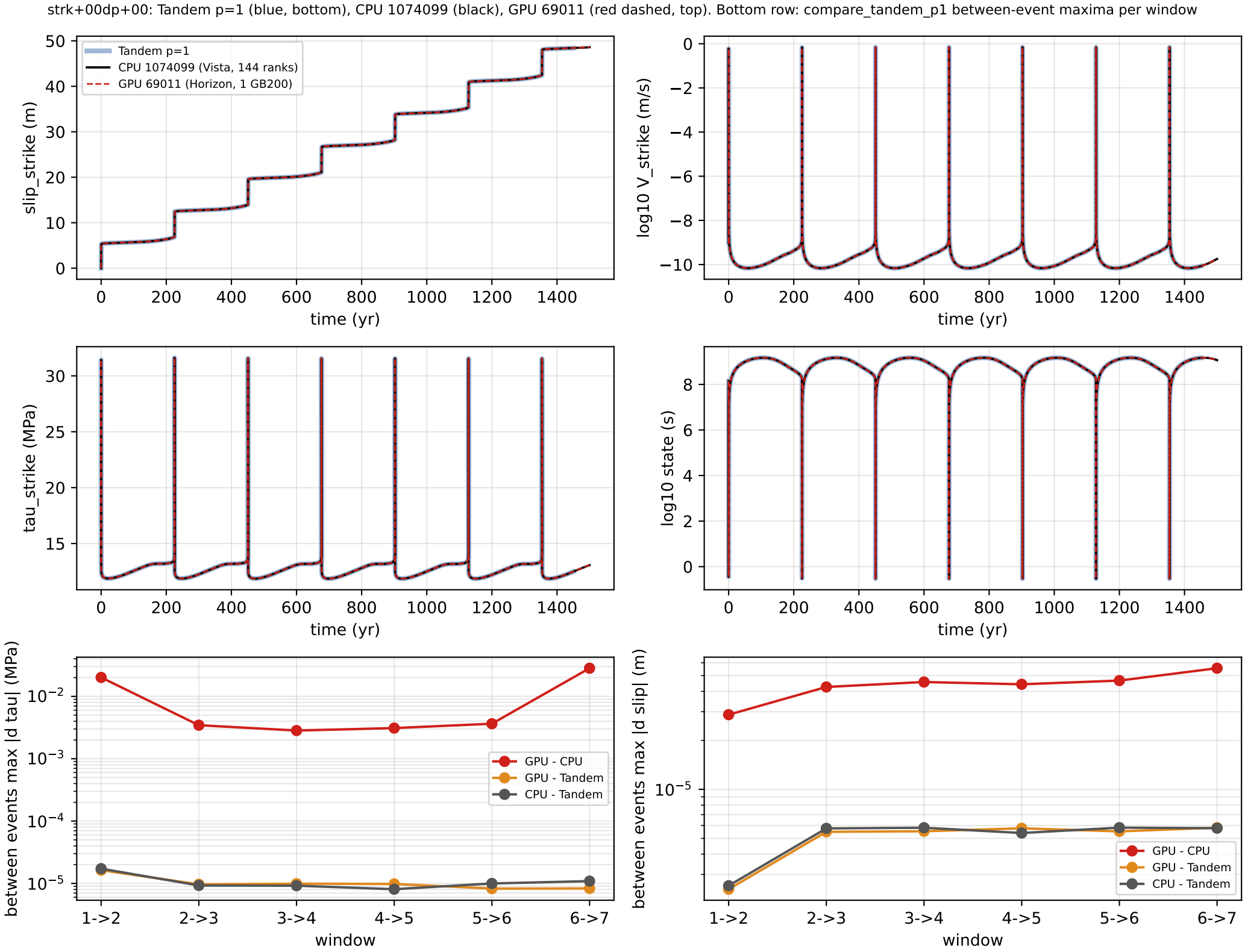

strk+00dp+10: Tandem p=1 (blue, bottom), CPU 1074099 (black), GPU 69011 (red dashed, top). Bottom row: compare_tandem_p1 between-event maxima per window
Tandem p=1
CPU 1074099 (Vista, 144 ranks)
GPU 69011 (Horizon, 1 GB200)
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
time (yr)
between events max |d tau| (MPa)
between events max |d slip| (m)
GPU - CPU
GPU - Tandem
CPU - Tandem
1->2
2->3
3->4
4->5
5->6
6->7
window

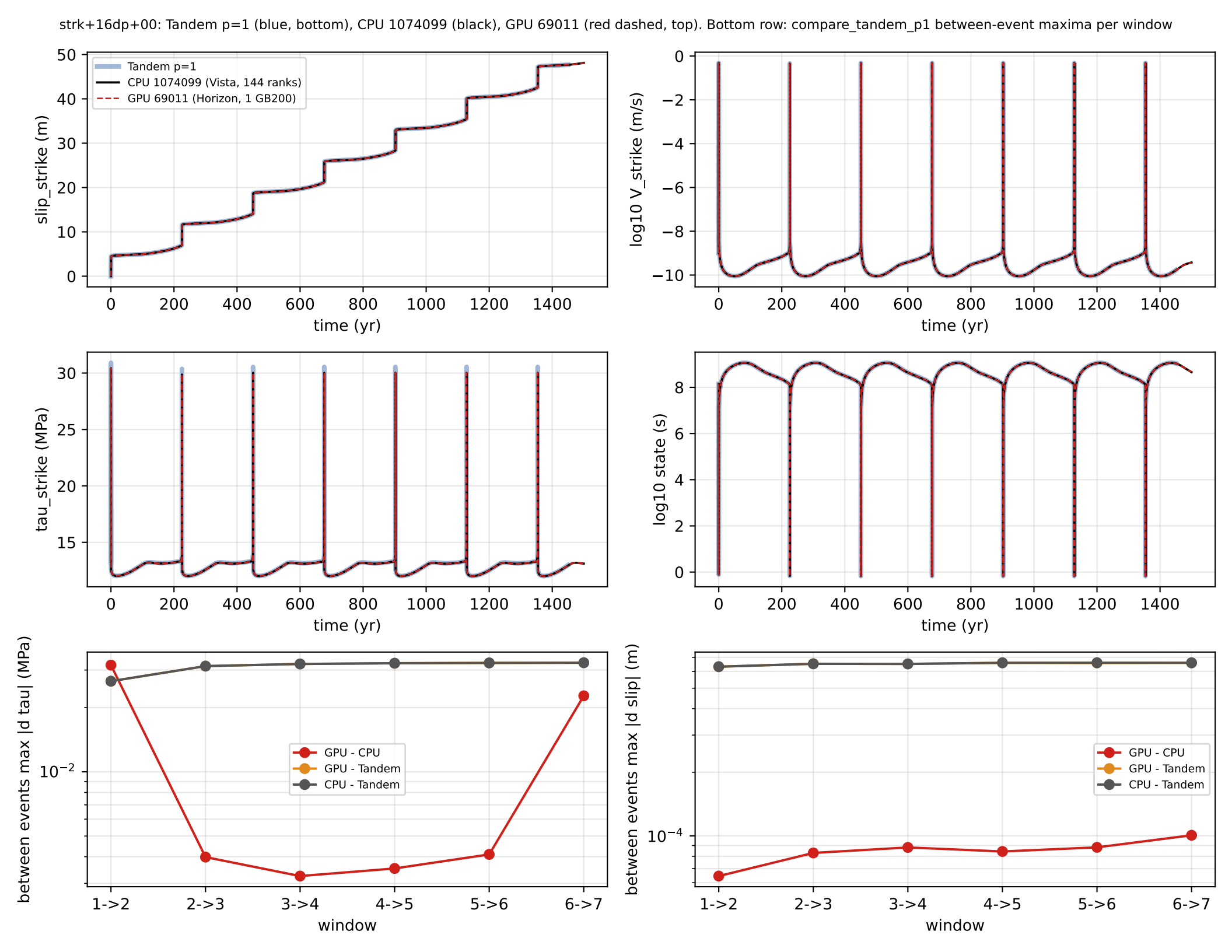

strk+16dp+10: Tandem p=1 (blue, bottom), CPU 1074099 (black), GPU 69011 (red dashed, top). Bottom row: compare_tandem_p1 between-event maxima per window
Tandem p=1
CPU 1074099 (Vista, 144 ranks)
GPU 69011 (Horizon, 1 GB200)
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
time (yr)
between events max |d tau| (MPa)
between events max |d slip| (m)
GPU - CPU
GPU - Tandem
CPU - Tandem
1->2
2->3
3->4
4->5
5->6
6->7
window

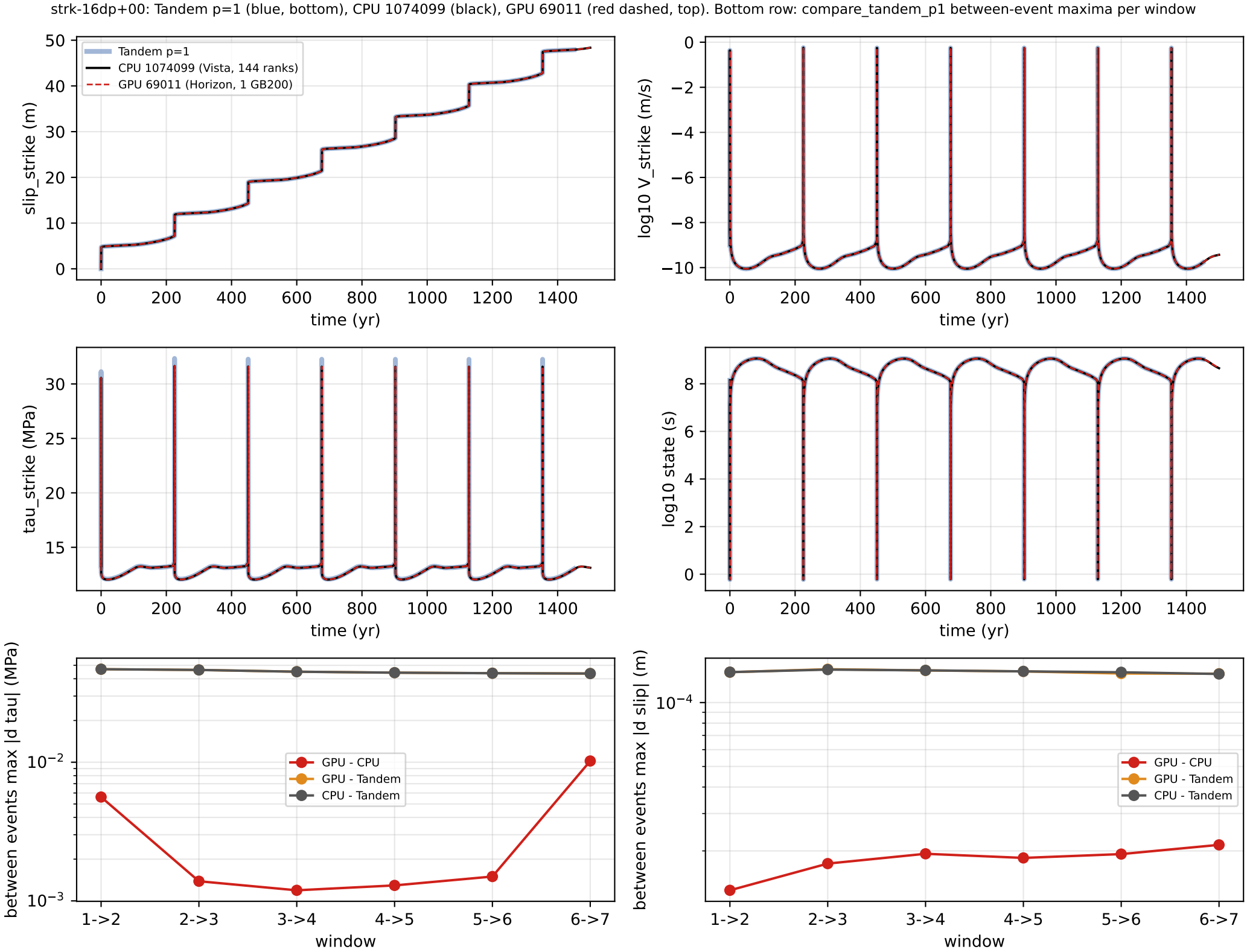

strk-16dp+10: Tandem p=1 (blue, bottom), CPU 1074099 (black), GPU 69011 (red dashed, top). Bottom row: compare_tandem_p1 between-event maxima per window
Tandem p=1
CPU 1074099 (Vista, 144 ranks)
GPU 69011 (Horizon, 1 GB200)
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
time (yr)
GPU - CPU
GPU - Tandem
CPU - Tandem
between events max |d tau| (MPa)
between events max |d slip| (m)
window
1->2
2->3
3->4
4->5
5->6
6->7

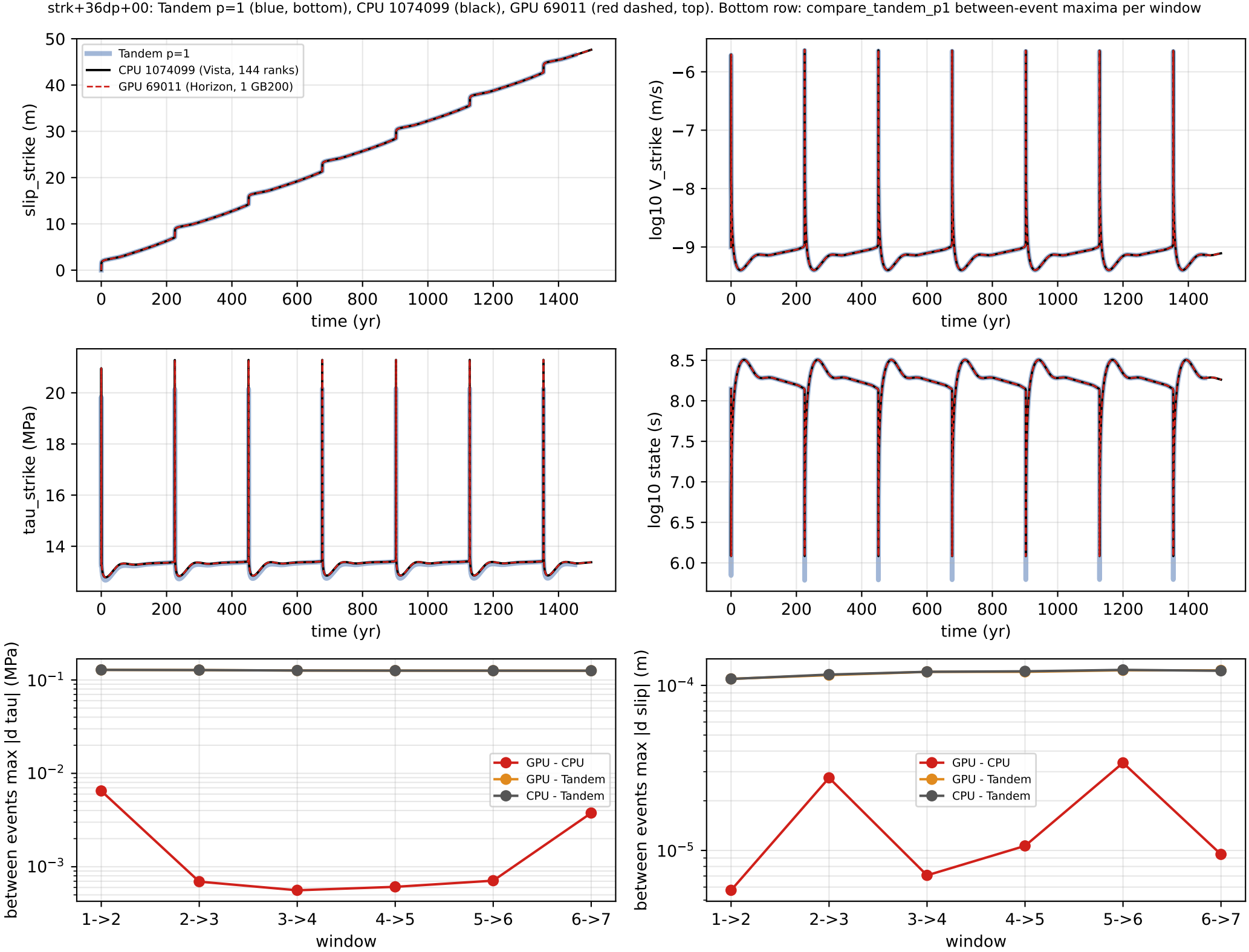

strk-24dp+10: Tandem p=1 (blue, bottom), CPU 1074099 (black), GPU 69011 (red dashed, top). Bottom row: compare_tandem_p1 between-event maxima per window
Tandem p=1
CPU 1074099 (Vista, 144 ranks)
GPU 69011 (Horizon, 1 GB200)
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
time (yr)
between events max |d tau| (MPa)
between events max |d slip| (m)
GPU - CPU
GPU - Tandem
CPU - Tandem
1->2
2->3
3->4
4->5
5->6
6->7
window

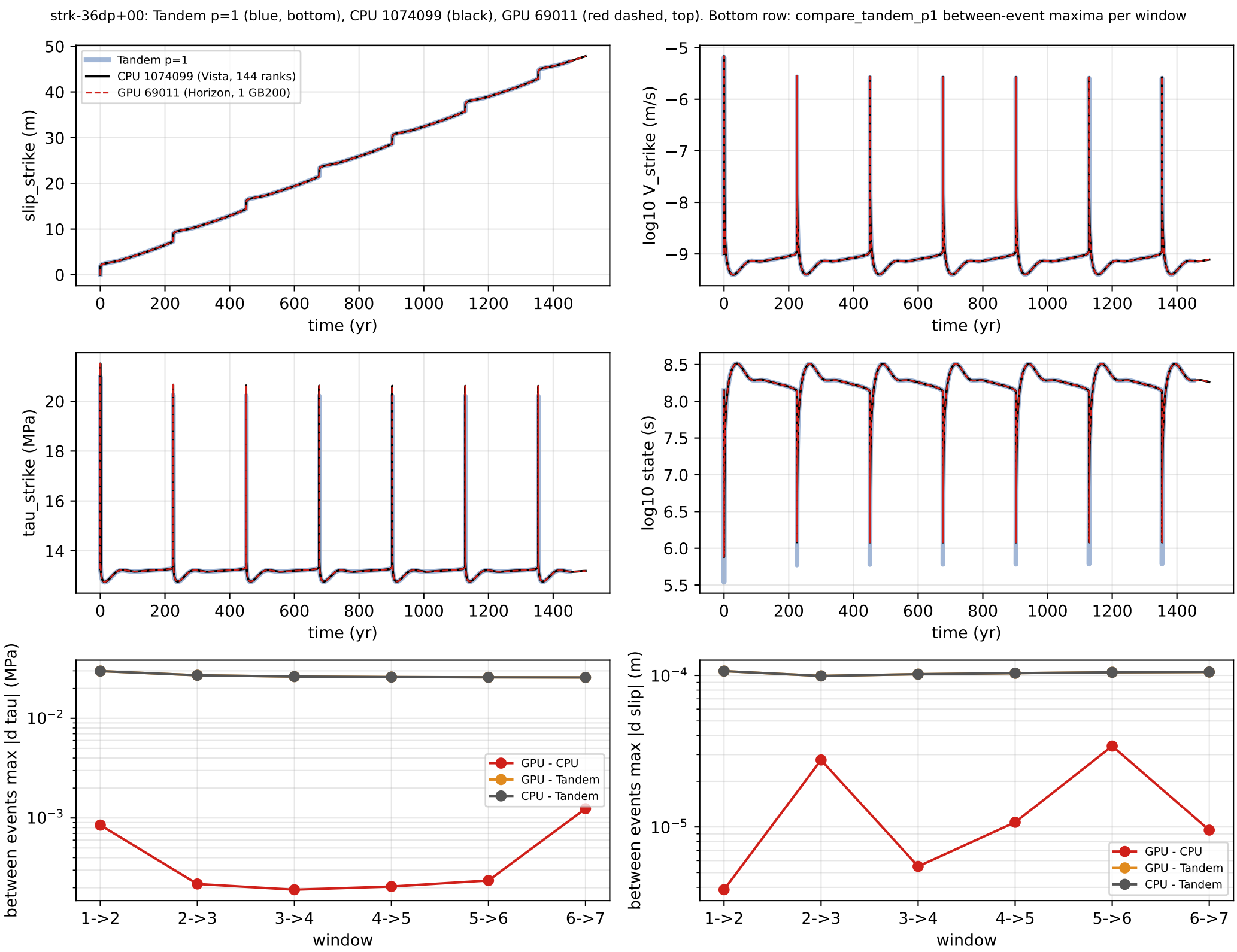

strk+00dp+22: Tandem p=1 (blue, bottom), CPU 1074099 (black), GPU 69011 (red dashed, top). Bottom row: compare_tandem_p1 between-event maxima per window
Tandem p=1
CPU 1074099 (Vista, 144 ranks)
GPU 69011 (Horizon, 1 GB200)
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
time (yr)
between events max |d tau| (MPa)
between events max |d slip| (m)
GPU - CPU
GPU - Tandem
CPU - Tandem
1->2
2->3
3->4
4->5
5->6
6->7
window


strk+00dp+00: Tandem p=1 (blue, bottom), CPU 1074099 (black), GPU 69011 (red dashed, top). Bottom row: compare_tandem_p1 between-event maxima per window
Tandem p=1
CPU 1074099 (Vista, 144 ranks)
GPU 69011 (Horizon, 1 GB200)
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
time (yr)
between events max |d tau| (MPa)
between events max |d slip| (m)
GPU - CPU
GPU - Tandem
CPU - Tandem
1->2
2->3
3->4
4->5
5->6
6->7
window


strk+16dp+00: Tandem p=1 (blue, bottom), CPU 1074099 (black), GPU 69011 (red dashed, top). Bottom row: compare_tandem_p1 between-event maxima per window
Tandem p=1
CPU 1074099 (Vista, 144 ranks)
GPU 69011 (Horizon, 1 GB200)
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
time (yr)
between events max |d tau| (MPa)
between events max |d slip| (m)
GPU - CPU
GPU - Tandem
CPU - Tandem
1->2
2->3
3->4
4->5
5->6
6->7
window


strk-16dp+00: Tandem p=1 (blue, bottom), CPU 1074099 (black), GPU 69011 (red dashed, top). Bottom row: compare_tandem_p1 between-event maxima per window
Tandem p=1
CPU 1074099 (Vista, 144 ranks)
GPU 69011 (Horizon, 1 GB200)
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
time (yr)
between events max |d tau| (MPa)
between events max |d slip| (m)
GPU - CPU
GPU - Tandem
CPU - Tandem
1->2
2->3
3->4
4->5
5->6
6->7
window


strk+36dp+00: Tandem p=1 (blue, bottom), CPU 1074099 (black), GPU 69011 (red dashed, top). Bottom row: compare_tandem_p1 between-event maxima per window
Tandem p=1
CPU 1074099 (Vista, 144 ranks)
GPU 69011 (Horizon, 1 GB200)
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
time (yr)
between events max |d tau| (MPa)
between events max |d slip| (m)
GPU - CPU
GPU - Tandem
CPU - Tandem
1->2
2->3
3->4
4->5
5->6
6->7
window


strk-36dp+00: Tandem p=1 (blue, bottom), CPU 1074099 (black), GPU 69011 (red dashed, top). Bottom row: compare_tandem_p1 between-event maxima per window
Tandem p=1
CPU 1074099 (Vista, 144 ranks)
GPU 69011 (Horizon, 1 GB200)
slip_strike (m)
log10 V_strike (m/s)
tau_strike (MPa)
log10 state (s)
time (yr)
between events max |d tau| (MPa)
between events max |d slip| (m)
GPU - CPU
GPU - Tandem
CPU - Tandem
1->2
2->3
3->4
4->5
5->6
6->7
window